Ayan Ahmed

Software Engineer, Google

H +91 9035212894
B monishaj.65@gmail.com
Í monisha-jega.githu
monisha-jega
monisha-jegadeesan

Education

2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78

2015 **XII - Karnataka Board,** *KLE Society's Independent PU College, Bangalore* 97.30 %

2013 **X - ICSE,** *B P Indian Public School, Bangalore* 96.33%

Professional Experience

Dec 2022 - Present
Software Engineer, Level IV, Google LLC, New York
- Working on Keep, a notetaking editor in Google Workspace.

Aug 2020 - Nov 2022
Software Engineer, Level IV, Google India Pvt Ltd, Bangalore
- Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
- Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
- Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
- Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.

May 2019 - July 2019
Software Engineering Intern, Google India Pvt Ltd, Bangalore
Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.

May 2018 - July 2018
Research Intern, Big Data Experience Labs, Adobe Research, Bangalore
Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

Sep 2019 - May 2020
Paraphrase Generation with a Bilingual Model and Continuous Embeddings *Master's Thesis,* ***Language Technologies Institute, Carnegie Mellon University***
Machinated a novel technique for paraphrase generation using the von Mises-Fisher (vMF) Loss on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by *Prof. Yulia Tsvetkov*.

May 2017 - July 2017
Cognitive Approach to Natural Language Processing
Research Intern, ***Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore***
Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by Prof. Veni Madhavan.

Publications and Patents

[Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan*, Sachin Kumar, John Wieting, Yulia Tsvetkov*
In **Workshop on Multilingual Representation Learning**,
The 2021 Conference on Empirical Methods in Natural Language Processing (**EMNLP 2021**)

[Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
In **ACM CODS-COMAD 2020** - Young Researchers' Symposium

[Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, ***Monisha Jegadeesan****, Amrit Singhal, Kush Kumar Singh, Kundan Krishna*
In ACM User Interface Software and Technology Symposium 2019 (**ACM UIST 2019**)

[Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, ***Monisha Jegadeesan****, Amrit Singhal, Kush Kumar Singh, Kundan Krishna*
Filed at the US PTO (Application Number: 16/247,235)

[Publication and Poster] **Leveraging Ontological Knowledge for Neural Language Models (Paper, Poster)** *Ameet Deshpande,* ***Monisha Jegadeesan***
In **ACM CODS-COMAD 2019** - Young Researchers' Symposium

Projects

Agentic AI-Driven Customer Support Chatbot (2024)

Designed and implemented a real-time AI-powered chatbot leveraging multi-agent orchestration for enterprise customer support. Developed a retrieval-augmented generation (RAG) pipeline with vector embedding for knowledge grounding and enhanced query accuracy. Translated ambiguous business requirements into clear use cases and agent responsibilities, defining human-in-the-loop versus autonomous execution strategies. Delivered scalable, cloud-native architecture utilizing LangChain for agent orchestration and OpenAI embeddings for document ingestion, reducing human workload by 35% and improving response latency to near real-time.

Tech: LangChain, OpenAI API, ChromaDB, Python, Vector Embeddings

Call Center Performance Analytics and Automation Blueprint (2024)

Led solution architecture for an end-to-end call center performance monitoring system, framing business problems into technical KPIs and prioritizing automation opportunities. Designed event-driven workflows integrating SQL data aggregation, real-time analytics, and Power BI visualization to enable agent performance and SLA compliance tracking. Created clear deliverables such as dashboard interaction flows, success metrics, and phased rollout plans. Resulted in optimized agent allocation, 22% reduction in wait time, and enhanced governance through role-based access and audit controls.

Tech: SQL, Power BI, DAX, Data Modeling, Event-Driven Architecture

Cloud-Native Sales Forecasting Platform with AI Integration (2024)

Architected a scalable AI-powered forecasting solution for retail sales demand, focusing on model selection trade-offs between accuracy and latency using ARIMA and Prophet. Defined phased delivery from MVP to autonomous optimization with structured phase gating. Developed data ingestion and preprocessing pipelines embedded within AWS Lambda-driven workflows to support cost-effective, event-driven execution. Delivered time series insights that reduced forecasting error by 25%, aligning business objectives with technical solution scope and risk controls.

Tech: Python, Prophet, ARIMA, Pandas, AWS Lambda, Event-Driven Architecture

Enterprise Traffic Analytics and Conversion Optimization Solution (2024)

Consulted and designed an end-to-end analytics platform for web traffic that prioritized automation and agentic workflows to enhance marketing ROI. Decomposed ambiguous conversion challenges into structured KPIs and agent responsibility matrices. Integrated SQL-based data querying with Power BI dashboards for real-time decision making, incorporating role-based access and cost monitoring. Delivered a phased roadmap and architecture pack supporting scalable, secure data retrieval and prompt versioning strategies, boosting conversion rates by 17%.

Tech: Google Analytics, SQL, Power BI, Data Visualization, Role-Based Access

About :

Experienced AI Solution Architect focused on designing secure, scalable, cloud-native agentic AI architectures leveraging AWS services including Lambda, IAM, S3, and Amazon Bedrock ecosystem. Skilled in consulting with stakeholders to translate business problems into clear technical solutions, defining multi-agent orchestration workflows, and producing delivery-ready solution artifacts. Proven expertise in LLM-based systems, prompt engineering at scale, and hybrid model strategies for cost, latency, and accuracy optimization. Adept at agent responsibility matrices, AI governance, risk mitigation, and event-driven architectures. Committed to delivering phased roadmaps, scope of work, and scalable AI automation platforms that drive enterprise-grade agentic AI solutions aligned with business objectives.